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Learning under ambiguity: An experimental investigation ☆

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ABSTRACT

We investigate learning in ambiguous situations where subjects bet on a winning event whose probability depends on an unknown proportion of winning chips in an urn. Varying the number of draws prior to choice allows us to "scan" ambiguity attitudes across differing amounts of information. By separately eliciting posterior beliefs in addition to matching probabilities, we disentangle the impact of learning on ambiguity attitude from its impact on beliefs, including divergences from Bayesian update. Both "raw data" and smooth ambiguity model-based analyses show that learning affects ambiguity attitude in the direction of ambiguity neutrality. Moreover, at small sample sizes, the impact of these changes on preferences is comparable to that of the divergence from Bayesian update.

1. Introduction

In classic examples, Ellsberg (1961) has shown that the standard model of choice under uncertainty, Savage's (1954) subjective expected utility (SEU), fails to capture ambiguity attitudes. In particular, people who prefer betting on an urn containing 50 black balls and 50 red balls to betting on an urn containing 100 black and red balls in an unknown proportion, no matter the winning color, display ambiguity aversion. Whilst suggestive, these situations are extreme compared to those encountered in everyday life. More often than not, decision makers are not given the probabilities of the realization of the relevant outcomes, but they are not totally ignorant of them either. They have had some opportunity for learning.

Standard Bayesianism tells a clear-cut story about choice under learning. Bayesian decision makers choose according to SEU, with a utility function that remains fixed in the face of new information, and beliefs – or subjective probabilities – that are updated according to Bayes rule. Since SEU decision makers are ambiguity neutral, there is no issue of whether their ambiguity attitudes change with learning. However, the same cannot be said for decision makers exhibiting a taste for or aversion to ambiguity, such as those making Ellsberg choices. For such decision makers, does learning affect *only* their beliefs, or does it *also* impact their ambiguity attitudes?

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This empirical question has been largely unexplored to date. The typical assumption would be that, as a taste, ambiguity aversion, like risk attitude, should remain fixed under learning. However the evidence on source dependence (Fox and Tversky, 1995) suggests another possibility. Source dependence features in several prominent ambiguity models, including source theory (Abdellaoui et al., 2011a) – where Ellsberg preferences are accommodated via source-dependent probability weighting à la Tversky and Fox (1995) – and second-order models, such as the smooth ambiguity model (Klibanoff et al., 2005; Nau, 2006; Ergin and Gul, 2009) – where the "ambiguity attitude" transformation function is often interpreted as reflecting the "different degrees of aversion to uncertainty across sources" (Marinacci, 2015, p1052). But since the quantity of information provided is typically assumed to determine a source of uncertainty (Tversky and Fox, 1995), the relevant source varies on learning, leaving space for the possibility that ambiguity attitudes evolve too. Whether learning impacts ambiguity attitudes is thus an open empirical question. This paper undertakes a systematic experimental study of it.

We adopt a simple yet flexible learning setup with urns containing 100 red and black chips in unknown proportions. Subjects receive evidence by sampling with replacement n times, where n reflects the amount of available information. After sampling, the ambiguous prospect $(x, E_{\tilde{p}})$ gives a monetary gain x if the "winning event" $E_{\tilde{p}}$ occurs, and nothing otherwise. The event $E_{\tilde{p}}$ corresponds to drawing a chip of the winning color from the sampled urn, with $\tilde{p}$ denoting the unknown proportion of such chips (e.g. of red chips). Specifically, we investigate the impact of learning on ambiguity attitudes in two experimental investigations: each involves a range of urn compositions, and hence proportions of winning chips $\tilde{p}$, with one focused on small samples ($n = 0, 5, 10$) and the other on large or very large ones ($n = 100, 10000$; see Table 3.1, Section 3). We thus cover the range between choice with no sampling – corresponding to Ellsberg's ambiguous urn – and choice after very large samples – which are closer to his risky urn.

A key challenge in learning contexts, which afflicts commonly used ambiguity aversion indices, is to disentangle ambiguity aversion from other behavioral factors, notably beliefs. Consider an ambiguous prospect $(x, E_{\tilde{p}})$ sampled n times. We elicit the *matching probability* $m_{\tilde{p},n}$ for which the subject is indifferent between $(x, E_{\tilde{p}})$ and the risky prospect $(x, m_{\tilde{p},n})$ with (known) probability $m_{\tilde{p},n}$. For instance, if the winning event is "the next chip drawn from the urn is black", then the corresponding matching probability is denoted by $m_{black,n}$. These matching probabilities reflect ambiguity attitudes. Indeed, following Schmeidler (1989), their deviation from complementarity – as measured by $1 - (m_{black,n} + m_{red,n})$ in our case – has been proposed as a "model-free" index of ambiguity aversion (e.g. Keppe and Weber, 1995; Baillon et al., 2018). However, in most decision models other than the neo-additive special case of Choquet expected utility (Chateauneuf et al., 2007), this index corresponds to some combination of the ambiguity aversion parameter with components representing beliefs (Baillon et al., 2019, Thm 16, Sections 5 & 7). So whilst we can track this ambiguity aversion index after learning samples of different sizes using our matching probability data, drawing stronger conclusions about ambiguity attitude requires separating it from beliefs.

To meet this challenge, we also implement a standalone choice-based elicitation of the subject's posterior probability distribution over the set of possible proportions of winning chips in a given urn – i.e., over the possible values of $\tilde{p}$. For this, we use an extension of Abdellaoui et al.'s (2011a) exchangeability method, without assuming SEU or any specific ambiguity model. Inspired by Jaffray's (1989) suggestion that ambiguity attitude is reflected in the divergence between a subject's subjective probability and her matching probability, a second "raw data" ambiguity aversion index is deduced from the best fitting line between matching and subjective probabilities (e.g. Dimmock et al., 2016, who infer subjective probabilities from symmetry). However, although our data allows us to track variations of this index with sampling, under decision models other than the neo-additive Choquet one, there is no guarantee that it reflects only the ambiguity attitude parameter, independently of beliefs, nor that it coincides with the previous index.

We thus complement these raw-data approaches with a model-based analysis, relying on one of the most prominent decision models implementing an explicit separation of beliefs and ambiguity attitudes: the smooth ambiguity model (Klibanoff et al., 2005). This model represents a subject's beliefs by a second-order distribution over the possible probabilities of the winning event, of the sort elicited in our experiment. Moreover, it involves a utility transformation function that has been shown to capture ambiguity attitude. Our elicitations of matching probabilities and subjects' posterior probability distributions allow us to estimate this "ambiguity attitude" transformation function after each sample size. Indeed, our standalone elicitation of second-order beliefs allows for an original way to estimate ambiguity attitude under the smooth ambiguity model, by plugging the elicited beliefs into the model to simply infer the aforementioned "ambiguity attitude" transformation function from matching probabilities. To our knowledge, this is the first direct choice-based econometric estimation of the transformation function involving an explicit and separate control for beliefs, and may be considered as a contribution in itself, of independent interest (see Section 5).

Our first finding, which emerges from both the raw-data and model-based approaches, is that, in certain ranges, there is a clear impact of sample size on ambiguity aversion: attitudes move in the direction of ambiguity neutrality as sample size increases. We find this effect at small sample sizes (e.g. increasing sample size from 5 to 10), with both aggregate- and individual-level estimates; latent class analysis under the smooth model indicates that around two-thirds of subjects exhibit decreasing ambiguity aversion on learning. By contrast, we find little significant change in ambiguity attitude at large sample sizes. To our knowledge, this is the first study to document an impact of learning on ambiguity aversion (see Section 5).

To gauge the behavioral significance of this ambiguity aversion change, we compare it to divergences from Bayes rule. There is an extensive literature on violations of Bayesian update and its economic importance (Benjamin, 2019), including experimental work in the sort of bag-and-chips setup used here. Drawing on our elicitation of posterior beliefs, we decompose the difference in matching probabilities between our subjects and "ideal" agents who have constant ambiguity aversion *and* update according to Bayes rule into the part due to ambiguity attitude change, and the part due to non-Bayesian update. At most prior probability levels, the former component dominates the latter in our data, suggesting that in our learning context, ambiguity aversion change has at least as large, if not larger impact on choice as violations of Bayes rule.

This paper is organized as follows. In Section 2 we set out the theoretical background, and in Section 3 we describe the experimental design. Section 4.1 presents the raw-data analyses in terms of the two aforementioned ambiguity aversion indices, whereas Section 4.2 reports the model-based analysis. Section 5 discusses related literature and outstanding issues.

2. Theoretical background

The present section first introduces preliminaries, including the general setup, notation and basic theoretical definitions. Then it introduces the smooth ambiguity model (Klibanoff et al., 2005). Finally it sets out the theory behind the treatment of beliefs in our studies.

2.1. Preliminaries

We consider decision-making situations where the objects of choice are two-outcome prospects that pay a monetary outcome $x > 0$ with a fixed winning probability p, and nothing otherwise. When the winning probability p is known to the decision maker from the outset, the prospect is said to be *risky* and we denote it by (x, p). In contrast, when the winning probability is fixed but unknown to the decision maker, e.g., the proportion of black chips in a given unknown urn containing 100 black and red chips, the prospect is *ambiguous* and denoted by $(x, E_{\tilde{p}})$, where $E_{\tilde{p}}$ stands for a winning event that has an unknown probability $\tilde{p} \in [0, 1]$. Before making a choice involving an ambiguous prospect $(x, E_{\tilde{p}})$, the decision maker samples with replacement from the underlying binomial probability distribution $n \geq 0$ times. The *matching probability* (henceforth MP) of the ambiguous prospect $(x, E_{\tilde{p}})$ that was sampled n times is the probability $m_{\tilde{p},n}$ such that $(x, m_{\tilde{p},n}) \sim (x, E_{\tilde{p}})$, where $\sim$ stands for indifference. Similarly, the matching probability corresponding to $(x, E^c_{\tilde{p}})$ is denoted by $m^c_{\tilde{p},n}$. For the sake of notational simplicity, the MPs of betting on "black" or "red" in the aforementioned (sampled) urn are denoted by $m_{black,n}$ and $m_{red,n}$, respectively.

2.2. Ambiguity aversion indices

As specified in the Introduction, our "raw-data" analysis will involve two ambiguity aversion indices. Following Schmeidler (1989), it has been recognized that non-additivity of matching probabilities for complementary events reveals a non-neutral attitude to ambiguity. Keppe and Weber (1995), for instance, report empirical evidence in Ellsberg-like experiments where the sum of the matching probabilities of an event and its complement is less than one, hence revealing ambiguity aversion. In our setup, the relevant difference is $1 - (m_{\tilde{p},n} + m^c_{\tilde{p},n})$; we call this the *complementarity ambiguity aversion index*. Elicitation of MPs of complementary events – the drawing of a red vs a black chip after sampling – for each sample size allows one to track how this index is impacted by learning.

Since the seminal work of Machina and Schmeidler (1992), it has been known that decision makers may exhibit preferences violating expected utility, although they are based on probabilistic beliefs, in the sense that there is a subjective probability for each event, and preferences over bets on events are entirely determined by this probability. By developing the notion of exchangeability, Chew and Sagi (2006) provide mild regularity and richness conditions for such *probabilistic sophistication* within families of events, i.e. conditions under which these events can be endowed with a subjective probability measure without committing to a specific preference functional on prospects. Under such conditions (which have been tested in subsequent work; Section 2.4), decision makers have well-defined subjective probabilities for events, which may differ from their corresponding MPs. And indeed, an alternative index of ambiguity aversion, initially mooted by Jaffray (1989), focuses on the relationship between a subject's MP for an event and her subjective probability for it. A MP which is higher (lower) that the corresponding subjective probability suggests ambiguity seeking (aversion). Ambiguity neutrality corresponds to the case of equality. Separate elicitation of matching probabilities and subjective probabilities after sampling would thus allow one to infer Jaffray-style indications of ambiguity attitude.

To implement such comparisons, we follow Dimmock et al. (2016, Section 3.1) and use the best-fitting line between MPs and the corresponding subjective probabilities – whether the latter are stipulated as in their experiment, or measured independently as in ours. Denoting the subjective probability by $\pi_{\tilde{p},n}$, let this line be $m_{\tilde{p},n} = c_n + s_n \pi_{\tilde{p},n}$, where c_n is the intercept and s_n the slope. Dimmock et al. define two "global ambiguity attitude indexes":

$$a_n = 1 - s_n \text{ and } b_n = 1 - s_n - 2c_n. \tag{1}$$

As b_n is inversely related to the average height of the line (elevation), they claim it measures ambiguity aversion. Index a_n reflects a lack of discrimination of intermediate levels of likelihood, i.e., likelihood insensitivity. Some recent contributions have related this index to ambiguity perception (Anantanasuwong et al., 2019; Baillon et al., 2018). Ambiguity aversion (likelihood insensitivity) increases as b_n (a_n) increases. Ambiguity neutrality corresponds to the case where $a_n = b_n = 0$. We call b_n the *belief-based ambiguity aversion index*.

The complementarity and belief-based ambiguity aversion indices, measured for the same subject, are not always the same. In fact, they are only guaranteed to be equal for subjects whose preferences adhere to the neo-additive capacity model of Chateauneuf et al. (2007). This belongs to the biseparable preference specification under probabilistic sophistication, in which preferences over prospects $(x, E_{\tilde{p}})$ are represented by[1]

[1] As standard, preferences are represented by a functional V on prospects, if, for any pair of prospects, the preferred prospect receives a higher value under V. Prospects which are indifferent receive the same V-value.

$$W_n(E_{\tilde{p}})u(x) + (1 - W_n(E_{\tilde{p}}))u(0), \tag{2}$$

The neo-additive capacity model is the special case where the willingness to bet on event $E_{\tilde{p}}$, $W_n(E_{\tilde{p}}) \in [0, 1]$, is a linear function of the subjective probability, for $\pi_{\tilde{p},n} \in (0, 1)$. For decision models other than this one, the value of the complementarity ambiguity aversion index will depend on the events used to estimate it (Baillon et al., 2021, Thm 16). Moreover, this index reflects some combination of the ambiguity parameter in the model and the belief representation Baillon et al. (2021, Sections 5 & 7). To this extent, analysis of learning in terms of these indices can only yield precise conclusions under the strong assumption of neo-additive capacities. For more robustness, we also analyze our data under an alternative preference model.

2.3. Smooth ambiguity

Under the smooth ambiguity model, preferences are represented by the functional assigning the following value to an ambiguous prospect $(x, E_{\tilde{p}})$ after having been sampled n times[2]:

$$\phi_n^{-1}\left(\int_{\Delta(S)} \phi_n\left(\lambda(E_{\tilde{p}})u(x) + (1 - \lambda(E_{\tilde{p}}))u(0)\right) d\mu_n(\lambda)\right), \tag{3}$$

where S is the underlying state space, μ_n is the posterior second-order probability distribution – a distribution over the space $\Delta(S)$ of probability distributions λ over S – $\phi_n : u(\mathbb{R}) \to \mathbb{R}$ is a strictly increasing function, and u is the usual von Neumann-Morgenstern utility function. As shown by Klibanoff et al. (2005), this model structurally separates the decision maker's beliefs – which in this model include perceived ambiguity and are represented by the second-order probability μ_n – from her ambiguity attitudes – captured by the transformation function ϕ_n. In the case of risk, μ_n is degenerate and the model coincides with expected utility.

The formal model (3) is flexible as concerns both ambiguity attitudes – permitting ambiguity aversion and ambiguity proneness, depending on the form of the transformation function ϕ_n – and the impact of learning on them. To our knowledge, it remains an open empirical question whether the transformation function varies across situations with differing "quantities of information"; our investigation will provide evidence on this question, by estimating ϕ_n for various sample sizes n.

Under this model then, the decision maker's full subjective beliefs correspond to a distribution μ_n over the set of probability distributions governing $E_{\tilde{p}}$ – that is, essentially, over the value of the underlying probability $\tilde{p}$.[3] More concretely, in our studies, where the winning event is the next chip drawn from an urn being of a certain color (Section 3), μ_n is a distribution over the possible compositions of the urn. For a fixed monetary gain x with normalized utilities $u(x) = 1$ and $u(0) = 0$, the previous equation thus reduces to the following evaluation of $(x, E_{\tilde{p}})$:

$$\phi_n^{-1}\left(\int_{[0,1]} \phi_n(p) d\mu_n(p)\right) \tag{4}$$

where the ambiguity attitude is captured by the transformation function ϕ_n, which may depend on the sample size. This expression yields the MP of event $E_{\tilde{p}}$ under the smooth model. It thus shows that the second-order belief distribution μ_n and MPs suffice to determine ϕ_n, with no need to estimate the utility u.

2.4. Beliefs

We elicit posterior beliefs from ex post choices, in the absence of any assumption about the nature of belief update and hence about how the ex post beliefs depend on the frequency of colors in the observed sample and its size. As explained in Sections 2.2 and 2.3, both the belief-based ambiguity aversion index and the smooth ambiguity model involve representations of beliefs which are probabilistic – in the case of the smooth model, at the second-order level. Our analysis and elicitation of beliefs for a given sample size will thus be probabilistic in nature. This allows us to tap into recent techniques developed for probabilistic belief elicitation under non-SEU preferences. More specifically, we elicit posterior beliefs without committing to any particular decision model or parametric form for beliefs at the outset. To do so, we employ a belief elicitation method that does not assume a specific preference functional under ambiguity.

We use a version of the exchangeability method of Abdellaoui et al. (2011a). The theory behind exchangeability, developed by Chew and Sagi (2006, 2008), shows that the conditions guaranteeing its success are precisely those guaranteeing probabilistic sophistication with respect to the relevant events (Section 2.2). Baillon (2008) and Abdellaoui et al. (2011a) have tested these conditions in several contexts, and have failed to reject them. Therefore, the basic method involves no assumptions beyond that of probabilistic beliefs, which is already involved in the decision approaches discussed above.

The winning event $E_{\tilde{p}}$ in our studies was realized by the draw of a chip of a certain color (red or black) from a 100-chip urn with a true proportion of red chips that was unknown to the subject. After having sampled from the urn n times, we elicited the posterior

[2] This formula presents what Marinacci (2015, Section 4.3) calls the "certainty equivalent measured in utils"; given the use of MPs in this paper, it is the most relevant here. Since ϕ_n is strictly increasing, this representation is behaviorally equivalent to one with the outer ϕ_n^{-1} omitted.

[3] The first-order distributions are Bernoulli distributions $B(1, \tilde{p})$, which, for ease of notation, we shall parametrize by $\tilde{p} \in [0, 1]$.

Table 3.1
General description of studies.

Studies	Treatments	Choice-based elicitations		Sampling	# Subjects
		Preferences	Beliefs		
S	$n = 0, 5, 10$	MPs	Exchangeability	Physical urn	$N = 80$
L	$n = 100, 10000$	MPs	Exchangeability	Virtual urn	$N = 60$

belief distribution μ_n over $[0, 1]$, the domain of the true proportion of red chips, through choices between prospects (x, I) yielding a positive monetary gain x if the proportion of red chips belongs to interval $I \subset [0, 1]$, and nothing otherwise. Two intervals I and J are said to be *exchangeable* if $(x, I) \sim (x, J)$. Under the aforementioned weak conditions (Chew and Sagi, 2008), exchangeable intervals have the same subjective probability, i.e. $\mu_n(I) = \mu_n(J)$. For a range of intervals $[\underline{r}, \bar{r}]$ (see Table 3.2b, Section 3), we obtained the partition of each interval into two exchangeable subintervals through the elicitation of the proportion of red chips r such that the subject was indifferent between prospects $(x, [\underline{r}, r))$ and $(x, [r, \bar{r}])$.[4] The resulting equations were used to deduce the parameters under parametric specifications for μ_n: as discussed in Section 2.5 and Appendix A.2, we consider several parametric specifications and retain that with the best fit.

The parameters of the elicited distribution are used to infer the probability the subject assigns to red on the next draw from the urn, as its mean

$$\pi_{red,n} = \int_{[0,1]} p(red) d\mu_n(p). \tag{5}$$

This determines the subjective probability of the winning event for a particular ambiguous prospect $(x, E_{\tilde{p}})$ concerning this urn: $\pi_{\tilde{p},n} = \pi_{red,n}$ if red is the winning color, and $\pi_{\tilde{p},n} = 1 - \pi_{red,n}$ otherwise. Under the smooth ambiguity model, the whole distribution μ_n is the relevant second-order probability distribution for Eq. (4).

2.5. Parametric specifications

We explore several parametric forms, and retain those with the best fit. For the beliefs, we compare the Beta, Binomial, Triangular and Truncated Normal distributions for the posterior subjective probability distribution over urn compositions.

For the smooth ambiguity transformation function ϕ_n, we consider the popular power (CARA) and exponential (CRRA) specifications, as well as the following normalized version of the expo-power specification initially proposed by Saha (1993):

$$\phi_n(x) = \frac{1 - e^{-\theta_n x^{\rho_n}}}{1 - e^{-\theta_n}} \tag{6}$$

with $\theta_n \neq 0, \rho_n > 0$[5] and x taking values on the utility scale $[0, 1]$ appropriate for risky two-outcome prospects under the utility normalization set out in Section 2.3. As explained by Klibanoff et al. (2005), the transformation function plays an analogous role to the utility function in decision under risk, and the flexible expo-power parametric form has proved popular in the literature on risk because of its ability to encompass a variety of risk attitudes. In particular, the CARA exponential specification is a special case ($\rho_n = 1$) and the CRRA power specification is a limiting case (as $\theta_n \to 0$; Peel and Zhang (2009)). The expo-power specification thus involves weaker assumptions about subjects' ambiguity attitudes. Moreover, and unlike the power and exponential families, this family includes S-shaped convex-then-concave functions. Such functions reflect ambiguity proneness on the lower end of the (expected utility) scale and ambiguity aversion towards the upper end of the scale. This means that they can represent the preferences of subjects who exhibit an ambiguity seeking attitude for unlikely events – for which the second-order distribution μ_n is concentrated on low first-order probabilities of the winning event – but exhibit ambiguity aversion for likely events – and μ_n centered on high probabilities of the winning event. Since such behavior has been found in other studies (e.g. Kocher et al., 2018), it is relevant to consider a family encompassing, though not implying it.

3. Experimental design

We elicited MPs and posterior subjective probabilities from choice for prospects with unknown winning probabilities, after various amounts of sampling: small samples in study S and large samples in study L (indicated in Table 3.1).

3.1. Subjects

140 subjects were recruited in total from Laboratoire d'Economie Expérimentale de Paris subject pool (Table 3.1). Their choices were collected through computer-based individual interviews that lasted about one hour in each study. Each individual interview

[4] The exchangeability method used in Abdellaoui et al. (2011a) would have required the determination of the proportion r_1 of red chips that divides $[0, 1]$ into equally likely events first. Then, proportions r_2 and r_3 should be elicited to partition each of the two intervals $[0, r_1)$ and $[r_1, 1]$ into two equally likely subintervals, respectively, and so on. This chained procedure may have resulted in possible error propagation.

[5] Under the normalization adopted here, this is the condition ensuring that ϕ_n is strictly increasing.

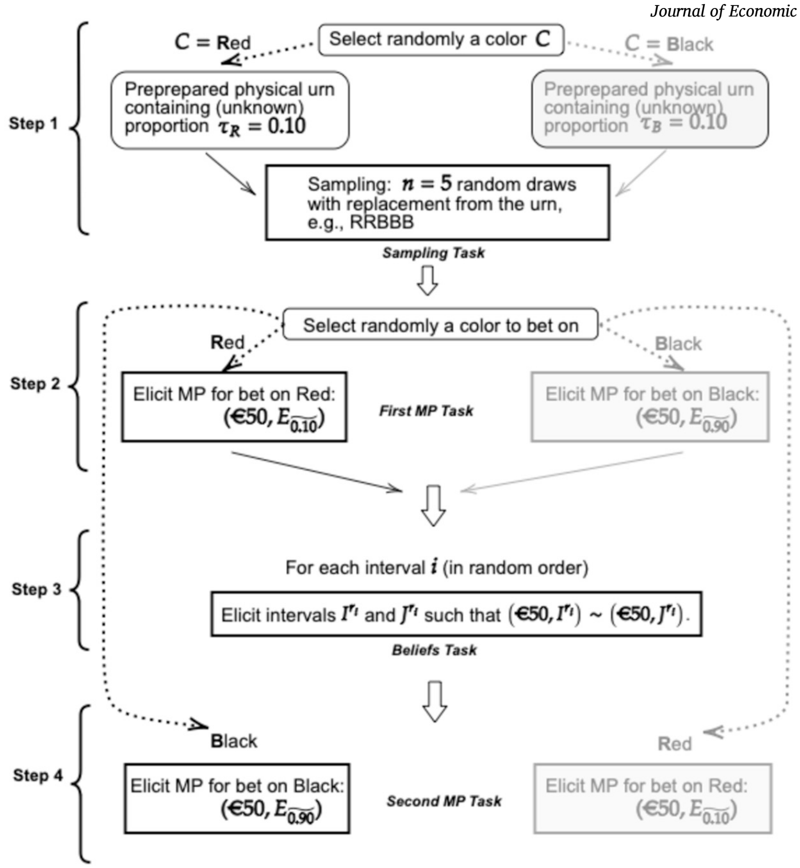

Fig. 3.1. Procedure of the typical experimental condition in study S (e.g., $n = 5, \tau_C = 0.10$). *Note:* Square boxes are tasks carried out by the subject, rounded boxes are randomizations performed by the experimenter.

started with a video presentation of the experimental instructions.[6] The information contained in the instructions, including prospect representation and sampling procedures, are reported in Appendix E. Subjects were told that there were no wrong or right answers, and that they could ask any question regarding the experiment.

3.2. Stimuli and treatments

MPs and beliefs in study S. The typical experimental condition in study S was characterized by a given sample size n and proportion of τ_C of chips of color C in the urn. Specifically, this study consisted of seven conditions: three conditions for each of the $n = 5, 10$ treatments, corresponding to three levels of the proportion τ_C, i.e., $0.1, 0.25, 0.50$; one for $n = 0$, involving the proportion $\tau_C = 0.3$. Each condition contains four steps corresponding to different tasks, which we now describe, using the case $n = 5$ and $\tau_C = 0.1$ for illustration; see Fig. 3.1 for a graphical representation.

1. In the first step (*sampling task*), a color $C =$ red or black was randomly selected. It was used to select the (preprepared) physical urn containing 100 chips, $100 \times \tau_C$ of which were of the color C, and the rest of which were of the other color (in our example, supposing that $C =$ red, the urn would contain 10 red chips and 90 black ones). The urn (whose composition was unknown to the subject) was presented to the subject for sampling purposes. The subject was asked to make n draws with replacement from the urn (in our example, 5 draws; e.g., RRBBB). The draws were recorded and remained visible on the computer screen throughout the rest of the condition.
2. In the second step (*first MP task*), a color (red or black) was randomly selected for betting purposes, and the subject's MP for a bet yielding €50 if the next chip drawn from the urn is that color, and nothing otherwise, was elicited. Note that the (unknown) underlying probability of winning depends on the composition of the urn and the color selected. Continuing with our example, if the urn contains 10 red chips, and the selected color was red, the bet is basically a bet on $E_{\widetilde{0.1}}$, and the subject was effectively answering a series of choice questions that provided the MP of $E_{\widetilde{0.1}}$. If black was selected, by contrast, then her responses indicate the MP of $E_{\widetilde{0.9}}$.

[6] The video presentations are available upon request.

Table 3.2
Stimuli for MPs and exchangeable events.

Unknown probability $\tilde{p}$ in prospect $(50, E_{\tilde{p}})$				
0.10	0.25	0.50	0.75	0.90

(a) Ambiguous prospects

Study	Event	Proportion to be assessed			
		r_1	r_2	r_3	r_4
S	I^{r_i}	$[0, r_1)$	$[0, r_2)$	$[0.50, r_3)$	$[0.25, r_4)$
	J^{r_i}	$[r_1, 1]$	$[r_2, 0.50]$	$[r_3, 1]$	$[r_4, 0.75]$
L	I^{r_i}	$[0, r_1)$	$[r^\star - r_2, r^\star + r_2]$		
	J^{r_i}	$[r_1, 1]$	$[0, r^\star - r_2) \cup (r^\star + r_2, 1]$		

Where $r^\star$ is the posterior probability of a red chip according to Bayesian update with a uniform prior (Section 2.4).

(b) Exchangeable events: $(50, I^{r_i}) \sim (50, J^{r_i})$

3. The third step (*beliefs task*) is devoted to the elicitation of the subject's beliefs regarding the proportion of red chips in the urn. Specifically, the subject answered a series of choice questions that elicited the four pairs of exchangeable intervals concerning the composition of the urn listed in Table 3.2b. The order of the four elicitations was randomized.
4. Like the second step, in the fourth step (*second MP task*) the subject answered a new series of choice questions to determine the MP of the bet on the color that was not selected in step 2, i.e. a bet on black if red was selected in step 2. As explained in that step, this is basically eliciting the MP for the complementary event to that determined in step 2, i.e. the MP of $E_{\widetilde{0.9}}$ if the MP of $E_{\widetilde{0.1}}$ was elicited in step 2.

Overall, the condition characterized by sample size $n > 0$ and proportion τ_C resulted in the elicitation of MPs of events $E_{\widetilde{\tau_C}}$ and $E_{\widetilde{\bar{\tau}_C}}$, where $\bar{\tau}_C = 1 - \tau_C$ (steps 2 and 4 in Fig. 3.1) after n draws. Additionally, the posterior subjective probabilities of these events were inferred from the second-order posterior distribution estimated from the four pairs of exchangeable intervals (step 3 in Fig. 3.1). All conditions involved independent draws from different urns, and subjects were aware of this. The treatment order was randomized, and the condition order was randomized within treatments.

MPs and beliefs in study L. The procedure in study L closely resembled that in Study S, but with an emphasis on large samples, $n = 100, 10000$. The only differences concern steps 1 and 3 of Fig. 3.1. Given the infeasibility of physical sampling for these sample sizes, the urns were realized on a computer. In step 1, the participant sampled from the virtual urn by pressing a button, allowing him to observe the cumulative results of the sequence of draws as they accumulated on the screen. In step 3, the difference concerned the intervals used in the exchangeability task. Given the large sample size, the subjective probability distribution μ_n was expected to be very concentrated around its mean, and the latter was expected to be close to the Bayesian prediction obtained from the update of a uniform prior, which we denote $r^\star$. Given the concentration, it seemed sufficient to elicit only the mean and variance of the distribution. More specifically, we partition $[0, 1]$ in two fashions (Table 3.2b). On the one hand, as in study S, we elicited r_1 such that intervals $[0, r_1)$ and $[r_1, 1]$ were exchangeable, hence obtaining the median of the distribution. On the other hand, we partitioned $[0, 1]$ into an interval centered around the Bayesian prediction $r^\star$, namely $[r^\star - r_2, r^\star + r_2]$, and its complement $[0, r^\star - r_2) \cup (r^\star + r_2, 1]$, through the elicitation of the r_2 for which the two sets are exchangeable. This gives an indication about the width of the distribution. We used the Bayesian prediction $r^\star$ as a proxy for the "middle" of the distribution, rather than the r_1 from the first elicitation, to ensure that the procedure was unchained, hence minimizing error propagation.

3.3. Elicitation techniques

MPs. In both studies, the MP of a given ambiguous prospect was elicited in a two-step procedure. First, a candidate MP was determined through a bisection process (Abdellaoui et al., 2008) that consisted in a series of single pairwise choices between the ambiguous prospect and a risky option. The bisection elicitation procedure aimed at circumventing a possible "middle-switching heuristic" effect, i.e. the tendency of subjects to switch in the middle of choice lists (Beauchamp et al., 2015). Then the complete "confirmation" scrollbar-based choice list, filled in according to the prior bisection choices, was displayed for verification (see Appendix E.3 for displays). The precision of the elicited MP was to the nearest 0.01. The confirmation step ensured that any errors during the bisection procedure could be corrected by the subject before validation. It was made clear in the instructions of the experiment that only the confirmation scrollbar-based choice list mattered for real incentives (e.g., Abdellaoui et al., 2023).

Exchangeable intervals. As concern beliefs, elicitation of pairs of exchangeable intervals proceeded as for MPs, using a bisection process followed by a single scrollbar-based choice list. Both involved choices between a bet $(50, I^r)$ yielding €50 when the proportion of red chips in the urn belonged to the interval I^r (and nothing otherwise), and a bet $(50, J^r)$ yielding €50 when the proportion of red chips in the urn belonged to the interval J^r.[7] The precision was 0.01: in the "confirmation" choice list, r (and hence the intervals I^r and J^r defined from it) was increased in steps of 0.01 (Appendix E.4).

3.4. Incentivizing subjects

Participants in the experiment received a flat payment of €10. Additionally, a random incentive system was implemented. For each study, a choice list was randomly chosen: it was either from the MP tasks related to the prospects reported in Table 3.2a or from the exchangeability tasks shown in Table 3.2b. Then, one single choice question from the selected choice list was randomly picked, the corresponding chosen option played out, and the resulting outcome paid in cash.

The use of physical urns in study S allowed us to apply the prior incentive system Prince (Johnson et al., 2021) according to which the choice question to be played out for real is randomly selected before the beginning of the experiment and only revealed at the end of it. Specifically, after the presentation of the instructions and before the beginning of the experiment, the subject randomly picked a chip from a bag containing numbered chips corresponding to the choice tasks the subject was to face. Then, she randomly picked a chip from another bag, corresponding to a question (among the possible 101 questions) on the "confirmation" choice list for MPs; she proceeded similarly for "confirmation" choice lists for exchangeable events (which were of different lengths). The subject was told that the draws determined the choice to be played for real, but that choice would be disclosed to her at the end of the experiment. The values were revealed to the subject at the end of the session, and the corresponding question on the appropriate "confirmation" choice list was played for real. If the chosen option involved the color of a chip drawn from an urn (MP question), a chip was physically drawn from this urn. If the chosen option concerned the composition of an unknown urn (exchangeability choice question), the composition (contents) of the urn used in the corresponding choice question was revealed to the subject. In both cases, the subject received the payoff according to her choice.

In study L, the Prince procedure could not be implemented because of the use of a computer-based virtual urn. We consequently opted for the standard random incentive mechanism, with a choice-list and a choice on it randomly selected and played for real at the end of the experiment. Both ambiguous and risky prospects were realized on computer.

4. Results

We now present our results concerning ambiguity attitudes in the presence of learning, while accounting for the effect of learning on beliefs. Basic statistics on the elicited beliefs are given in Appendix B.1; we focus here on the findings regarding ambiguity attitudes. We analyze the impact of learning on ambiguity attitudes without assuming a specific ambiguity model, in terms of the two ambiguity aversion indices introduced in Section 2.2 (Section 4.1). Then we investigate ambiguity attitude under the smooth ambiguity model. By comparing the estimated ambiguity-aversion transformation function across sample sizes, we evaluate how it changes with learning. Moreover, by combining the estimated ambiguity components with our data on posterior beliefs, we gain insight into the relative importance for choice behavior of the change in ambiguity aversion as compared to the deviation from Bayesian update (Section 4.2).

4.1. Ambiguity-model-free analyses

We first analyze our data using the two ambiguity aversion indices proposed in the literature (Section 2.2), which can be calculated without estimating a specific decision model.

Ambiguity attitude from matching probabilities. Fig. 4.1 draws box plots of the average complementarity ambiguity index over all urns, for each of the sample sizes in studies S and L (see Table A.6 for the corresponding empirical distributions of the complementarity index). Note that we recover the standard finding of moderate ambiguity aversion in the (two-color) Ellsberg unknown urn, which corresponds to our $n = 0$ case (Abdellaoui et al., 2023). It shows that in study S the median value of the index declines from 0.21 in the absence of sampling to 0.10 for $n = 5$ on the one hand, and from 0.10 to 0.05 for $n = 10$. Consistent with this pattern, the assumption of constant ambiguity aversion according to this index across sample sizes $n = 0, 5, 10$ is rejected (one-factor ANOVA with repeated measures, $p < 0.001$). Further, as illustrated in the Figure, paired t-tests reject equal ambiguity aversion between sample sizes 0 and 5 on the one hand ($p < 0.001$), and sample sizes 5 and 10 on the other hand ($p < 0.001$). At the individual level, 61 out of 80 subjects exhibited a higher ambiguity aversion for $n = 0$ than for $n = 5$ (Binomial, $p < 0.001$). Similarly, 54 out of 80 subjects exhibited a lower ambiguity aversion when they sampled 10 times than when they sampled 5 times (Binomial, $p < 0.001$).

As for study L, Fig. 4.1 shows a decline of median complementarity ambiguity aversion from 0.04 to 0.01 when the sample size increases from 100 to 10000. The assumption of equal ambiguity aversion across samples sizes is rejected by a paired t-test ($p < 0.001$). Additionally, 46 out of 60 subjects exhibited a higher index of ambiguity aversion for $n = 100$ than for $n = 10000$.

Moreover, note that the median ambiguity aversion index is positive, even at very large sample sizes ($n = 10000$); it is larger than 0 for 43 subjects out of 60 (Binomial, $p < 0.001$). This may suggest that even after extensive learning, attitudes toward ambiguity approach but do not fully reach the ambiguity-neutral benchmark of decisions made with known probabilities.

[7] It was made clear to subjects that the choices could be equivalently framed in terms of proportions of red or black chips. See Appendix E.4.

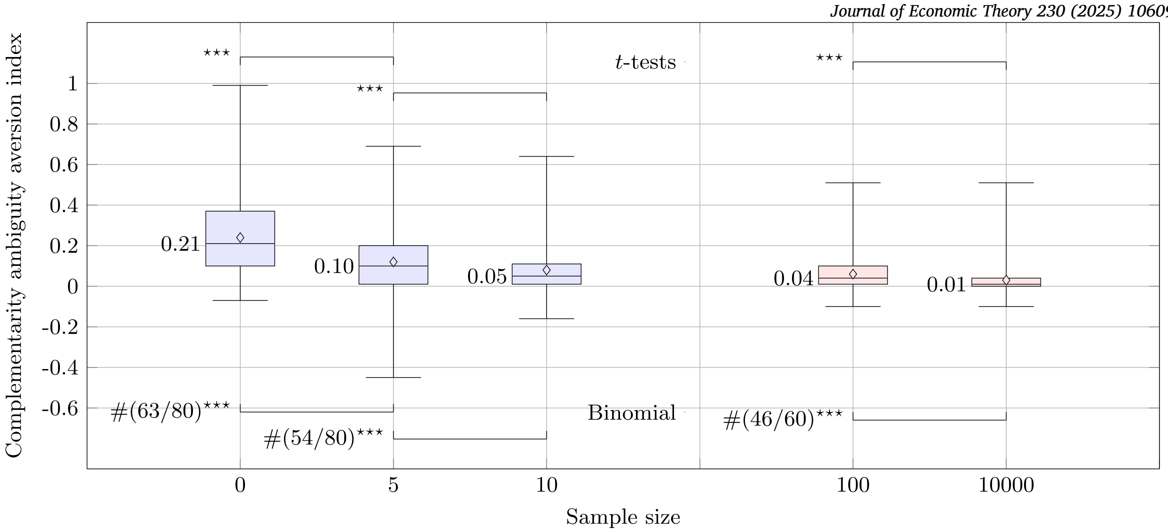

Fig. 4.1. Ambiguity aversion from MPs for sample sizes in studies S and L (Box plots). *Note.* In study S, the hypothesis of equal ambiguity aversion is rejected when n increases from 0 to 5 on the one hand, and from $n = 5$ to $n = 10$ on the other hand. Overall, ambiguity aversion declines when sample size increases. Although the decrease in ambiguity aversion is tiny is study L, the difference in ambiguity aversion is significant both at the aggregate and individual levels.

Table 4.1
ANCOVA regression: MPs vs. elicited posterior probabilities and sample size.

$m_{\tilde{p},n} = c_n + \Delta c \mathbf{1}_{n'} + (s_n + \Delta s \mathbf{1}_{n'})\pi_{\tilde{p},n} + \epsilon$		
Coefficients	Study S	
	$n = 5, n' = 10$	
	Estimates	Std. Error
Slope: s_n	0.782***	0.024
$\Delta c = c_{n'} - c_n$	−0.043***	0.013
$\Delta s = s_{n'} - s_n$	0.127***	0.028
Intercept: c_n	0.049***	0.011
R^2	0.840	
F-value	$F_{\mathrm{df}=(3,956)} = 1402.80$***	

*** : $p < 0.001$; ** : $p < 0.01$; * : $p < 0.05$;
ns : non-significant;
$\mathbf{1}_{n'} = 1$ if $n = n'$, and 0 otherwise; ϵ stands for a random error.

Ambiguity attitude from subjective vs. matching probabilities. We now consider the belief-based ambiguity aversion index. Since it requires a regression for calculation (Section 2.2), it cannot be estimated from our data on the $n = 0$ case. Nevertheless, our aggregate results in this case indicate beliefs and ambiguity attitudes in line with standard findings in the literature. On the one hand, the elicited subjective probability distribution exhibits symmetry around 0.5, as would be expected in the absence of information (we find Beta with $\alpha_0 = 1.33$ and $\beta_0 = 1.34$; Appendix A.1, Table A.1 and Appendix B.1). On the other hand, the empirical distributions of MPs point to average values that are clearly smaller than the corresponding subjective probabilities: the mean MPs are around 0.37 and 0.38 respectively, consistent with ambiguity aversion in the style of Ellsberg (see Tables A.1 and A.3 in Appendix A.1).

As concerns the other sample sizes, we first check whether the fitting lines for our data are affected by sample size at the aggregate level. One-way ANCOVA analysis shows that in study S elicited subjective probability (covariate), sample size (factor), and their interaction affect MPs ($p < 0.001$ for the three effects). Consequently, the regression lines of $m_{\tilde{p}}$ on $\pi_{\tilde{p}}$ corresponding to conditions $n = 5$ and $n = 10$ differ both in terms of intercept and slope. Table 4.1 reports the corresponding regression model estimates. We observe a decrease in the intercept ($\Delta c = c_{10} - c_5 < 0$) and a parallel increase in the slope ($\Delta s = s_{10} - s_5 > 0$) of the regression line when sample size increases. In summary, the aggregate analysis suggests an impact of sample size on the fitting line – and in particular on the belief-based ambiguity aversion index – at small sample sizes.

Fig. 4.2 reports box plots describing the empirical distributions of the belief-based likelihood insensitivity and ambiguity aversion indices estimated at the level of individual subjects (see also Table B.4, Appendix B.2). We observe that the median likelihood insensitivity (the horizontal lines in the box plots) is higher for $n = 5$ than for $n = 10$ (Fig. 4.2, left panel; paired t-test $p < 0.01$). The situation is similar for the belief-based ambiguity aversion index (b): the median value of this index is higher for $n = 5$ than for $n = 10$ (paired t-test, $p < 0.01$). At the individual level, the value of the ambiguity aversion index is higher in the $n = 5$ condition for 54 out

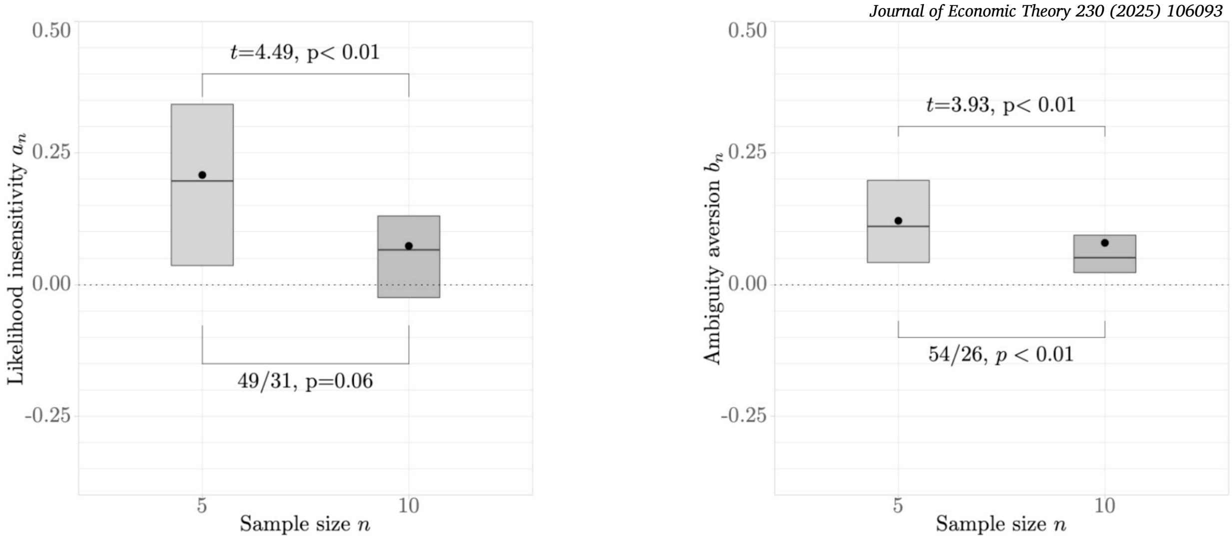

Fig. 4.2. Likelihood insensitivity and ambiguity aversion. *Note.* The hypothesis of an invariant insensitivity to probabilities across treatments is rejected at the aggregate level, but not at the individual level. On the other hand, ambiguity aversion varies across treatments both at the aggregate and individual levels.

of 80 subjects (Binomial, $p < 0.01$). Recalling that the ambiguity neutral benchmark is $a_n = b_n = 0$, both parameters move towards ambiguity neutrality on learning. To the extent that likelihood insensitivity is often considered a cognitive dimension, comparable with ambiguity perception or belief (Wakker, 2004), it is hardly surprising that it changes with learning. The (belief-based) ambiguity aversion index, by contrast, is typically interpreted as reflecting tastes; the finding that it is impacted by learning is more striking. It corroborates the message from our analysis of the complementarity ambiguity aversion index.

For study L, the picture regarding the impact of learning on the fitting lines is not as pronounced as in study S. Specifically, an ANCOVA analysis shows that the covariate (subjective probability) and the factor (sample size) affect matching probabilities ($p < 0.01$ in both cases), but their interaction barely has an impact ($p = 0.07$). This suggests different intercepts and identical slopes of the regression lines between conditions $n = 100$ and $n = 10000$. Regression analysis (provided in Table B.3, Appendix B.2) detects a significant, though very small, difference in intercepts. These results point to a constant perception of ambiguity across large samples on the one hand, and a tiny change in the belief-based ambiguity aversion index when samples size increases from 100 to 10000. The belief-based ambiguity aversion index exhibits a significant difference according to sample size (paired t-test, $p < 0.01$; Table B.4, Appendix B.2). At the individual level, it is higher when $n = 100$ for 46 vs. 14 subjects (sign-test, $p < 0.01$).

Moreover, a t-test shows that, in the large sample condition ($n = 10000$), the ambiguity aversion index differs from the "ambiguity neutral" benchmark ($H_0 : b_n = 0$, $p < 0.001$). Consistently with the analysis under the complementarity ambiguity aversion index, this could suggest a "residual gap" between the ambiguity attitude after having observed very large samples and the ambiguity neutral benchmark (Section 5).

4.2. *Analysis under the smooth ambiguity model*

We now analyze our data under smooth ambiguity preferences (Klibanoff et al., 2005). For the sake of generality, we report results on the "ambiguity attitude" transformation function ϕ_n under the expo-power parametrization throughout this section. In addition to the fact that this parametric form generalizes the power and exponential specifications, it also performs significantly better in terms of goodness of fit (Appendix A.2).

Small sample sizes: ambiguity attitudes. We first check whether ϕ_n is affected by sample size at the aggregate level, using a maximum likelihood procedure (Appendix D) with the expo-power specification of ϕ_n under the structural equation given in Section 2.3 (Eq. (4)). A likelihood ratio test clearly rejects the model with a single transformation function for both sample sizes in favor of two different ϕ_n for $n = 5$ and $n = 10$ ($p < 0.001$). This suggests that additional sampling affects ambiguity attitude when subjects still face a large amount of ambiguity. In order to refine this result, we introduced dummy variables capturing the possible changes in ϕ parameters ρ and θ between $n = 5$ and $n = 10$. Results (Appendix C, Table C.1) indicate statistically-significant effects on these two parameters, pointing towards a reduction in the curvature of ϕ when n increases.

We also conduct a latent class analysis of the impact of sample size on ϕ_n. Specifically, we consider two classes of subjects. A first class, representing a share λ of the sample, consists of subjects with a sample-size-independent ambiguity attitude (i.e. $\phi_5 = \phi_{10}$). A second class, representing a share $1 - \lambda$ of the sample, contains subjects for which ϕ_n depends on n (i.e. $\phi_5 \neq \phi_{10}$). That resulted in a likelihood estimation of the smooth model with an additional parameter, λ, to estimate (Appendix D and Table C.3). The results point to a non-negligible minority of subjects, $\lambda = 0.34$, exhibiting sample-size-invariant ambiguity attitude, with a solid majority of subjects having changed their ambiguity attitude when sample size was increased from $n = 5$ to $n = 10$.

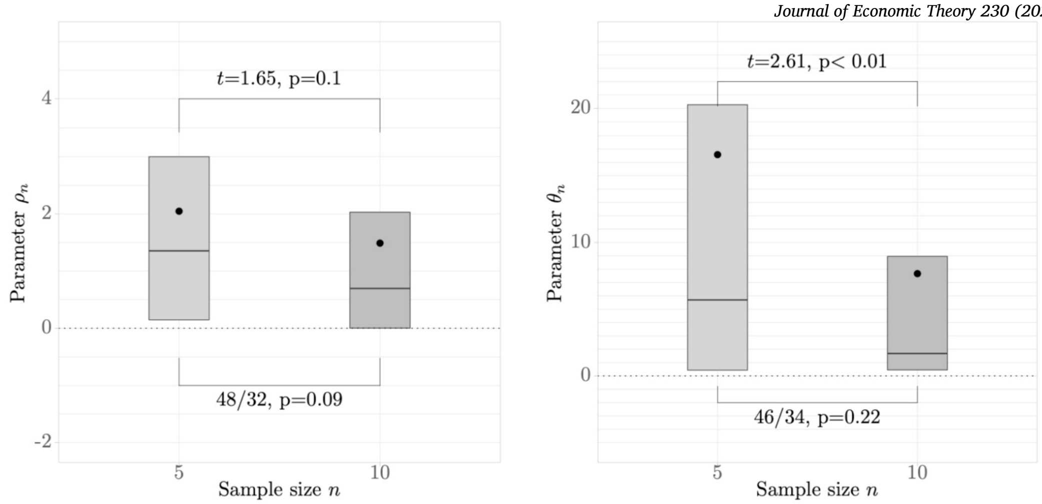

Fig. 4.3. Empirical distributions of the expo-power specification of ϕ_n. *Note.* The equality $\rho_5 = \rho_{10}$ is not rejected at both the aggregate and individual levels. Conversely, the equality $\theta_5 = \theta_{10}$ is rejected at the aggregate level (revealing a different concavity of ϕ_n across treatments), but not at the individual level.

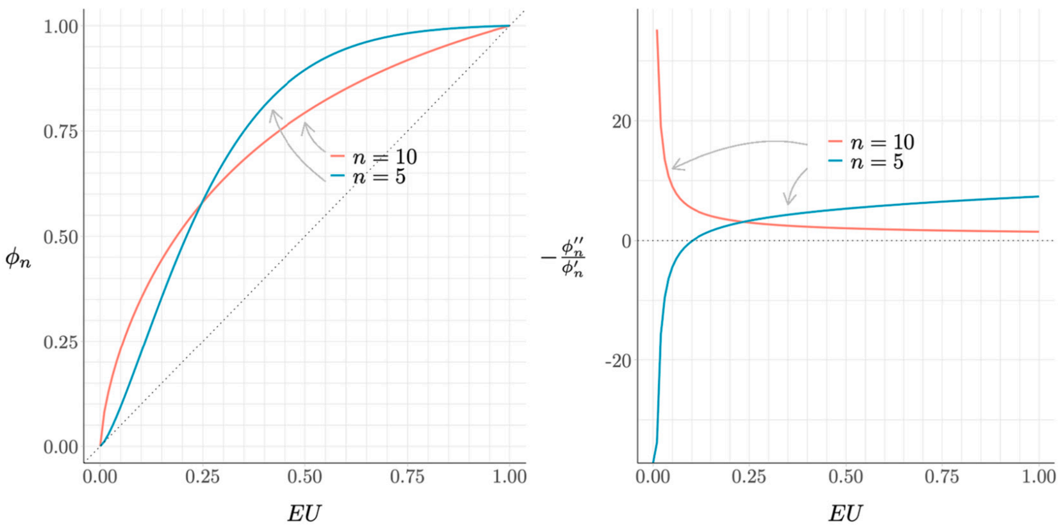

Fig. 4.4. Ambiguity attitude for median parameter estimates in study S. *Note.* At the middle and the top of the expected utility scale in the right hand panel, subjects exhibit ambiguity aversion, but are more ambiguity loving for the smaller sample size.

We also use a maximum likelihood procedure to estimate the parameters of the expo-power specification of ϕ_n at the individual level (see Appendix D for details). Fig. 4.3 draws means, medians and IQRs of the empirical distributions of parameters θ_n and ρ_n (numerical details are given in Table C.2, Appendix C). A paired t-test detects a significant difference in the θ_n parameter between the $n = 5$ and $n = 10$ treatments (Fig. 4.3, right panel). Moreover, Fig. 4.3 suggests that θ_n declines with sample size; this decline is also confirmed by the regression analysis reported in Table C.1 (Appendix C). As for parameter ρ, our observations are consistent with equal ρ_n for $n = 5$ and $n = 10$ (paired t-test, $p = 0.10$).

At the individual level, although a majority of subjects exhibit a decrease in θ_n on the one hand (46 subjects out of 80), and a decrease in ρ_n on the other (48 subjects out of 80), the corresponding sign tests are not significant at the 5% level. These results align with the latent class analysis, where a notable minority of subjects exhibit the same transformation function across treatments, hence explaining why the effect of sampling on ϕ_n is significant only at the aggregate level but not at the individual level.

Fig. 4.4 (left panel) plots the transformation functions ϕ_5 and ϕ_{10} for median estimates of the expo-power parameters in study S. Recall the analogy between the transformation function and the utility function in decision under risk: in particular, as Klibanoff et al. (2005) show, the ambiguity attitude is revealed by the curvature of ϕ_n. These authors have defined a "coefficient of ambiguity aversion", which is the ambiguity analogue of the Arrow-Pratt coefficient of (absolute) risk aversion; to bring out the relevant comparisons of ambiguity attitude, the right panel of Fig. 4.4 plots these coefficients for the transformation functions illustrated in the

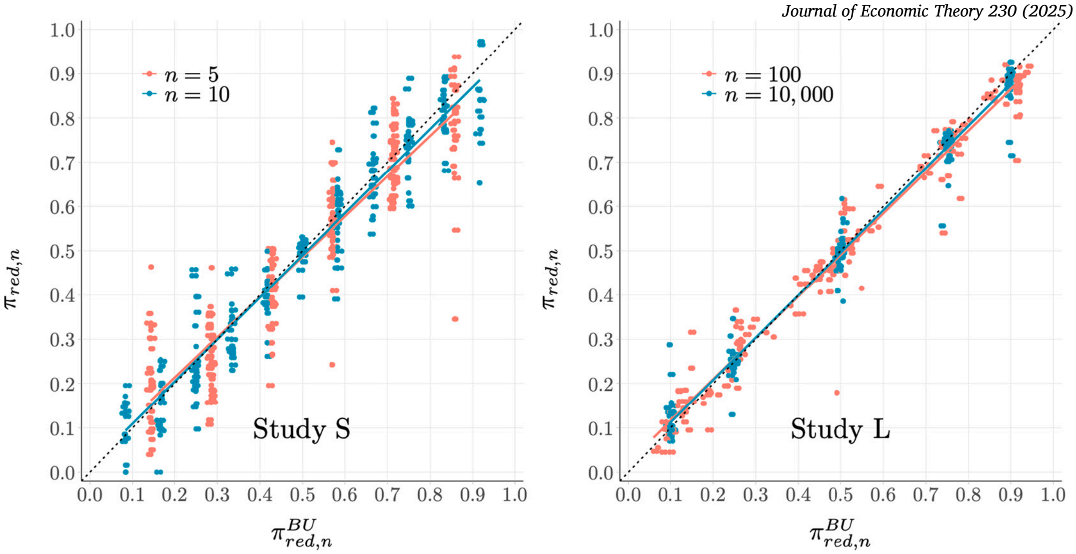

Fig. 4.5. Estimated posterior means vs. Bayesian update means. *Note.* Data points correspond to 80 subjects and 3×2 urns for study S and 60 subjects and 3×2 urns for study L (see Table 3.1 and Section 3.2).

left panel.[8] At the middle and top of the (expected utility) scale, subjects exhibit ambiguity aversion (the coefficient of ambiguity aversion is positive), but are more ambiguity averse for the smaller sample size $n = 5$. Indeed, t-tests reject the hypothesis of equal coefficients of ambiguity aversion for (utility) levels $EU = 0.5$ and $EU = 0.9$ ($p < 0.01$ in both cases). So for likely winning events, for which the second-order distribution μ_n is more centered on the top of the scale, subjects exhibit ambiguity aversion, which is stronger for the smaller sample size. For unlikely winning events – and hence μ_n concentrated near the bottom of the scale – there may be a tendency towards an ambiguity seeking attitude (i.e. negative coefficient of ambiguity aversion), but this is once again more pronounced for the smaller sample size. In general, over most of the scale, the finding of a change in ambiguity attitude on learning, in the direction of ambiguity neutrality, is upheld.

Ambiguity attitudes in small sample sizes: Summary. In summary, our data allows for comparison of ambiguity attitudes across sample sizes, both in terms of indices that do not assume a specific decision model and under the smooth ambiguity model, which permits a clean separation of ambiguity attitudes from beliefs. Under each analysis, we find that ambiguity attitudes – however they are operationalized – change with learning. Typically attitudes move towards ambiguity neutrality for larger samples.

Small sample sizes: ambiguity attitudes and belief update. The previous analysis reveals an effect of learning on ambiguity attitude. To ascertain its importance, we now compare it to the impact of divergence from Bayesian update.

This divergence can be defined by comparing the estimated beliefs after sampling with the posterior beliefs of a proper Bayesian, i.e. the posterior probability distribution resulting from the Bayesian update of an uninformative prior on the observed sample. More specifically, the Bayesian posterior for an observed sample of size n with a frequency f of red is the update of the prior uniform distribution over the probability interval $[0, 1]$, which is Beta$(1, 1)$, yielding[9]:

$$\text{Beta}(1 + fn, 1 + (1 - f)n). \tag{7}$$

This is the distribution over possible urn compositions (values of $\tilde{p}$) that a Bayesian would obtain with the same evidence; we denote it by μ_n^{BU}. Just as the estimated distribution μ_n generates a probability of red in a draw from the urn, $\pi_{red,n}$, as its mean (Section 2.4), the Bayesian posterior generates a Bayesian probability of red $\pi^{BU}_{red,n}$. For a prospect $(x, E_{\tilde{p}})$ after n samples, the Bayesian probability of winning is thus $\pi^{BU}_{\tilde{p},n}$.

As an illustration of the extent of divergence from Bayesian update as concerns these means, Fig. 4.5 draws the regression lines for $\pi_{red,n}$ against $\pi^{BU}_{red,n}$. It suggests a slight divergence of subjects' beliefs compared to Bayesian update. This accords with Fisher tests

8 Note that the expo-power specification has the property that this coefficient tends to $\pm\infty$ as $x \to 0$; the coefficient for small expected utility values in the graphs should be interpreted with this in mind, though this fact should not undermine any conclusions drawn at higher expected utility values. For this reason, we also do not perform tests at low expected utility values.

9 Since the typical ambiguous prospect in our experiments is generated by n draws from a binomial distribution with an unknown winning probability, i.e. $B(1, \tilde{p})$, standard Bayesian practice is to work with the Beta distribution over $\tilde{p}$. This distribution is known to be suitable in binomial settings where the winning probability is unknown to the decision maker. Moreover, beyond being the conjugate prior for the binomial distribution, the Beta distribution supports versatile modeling of unknown proportions (Berry, 1996).

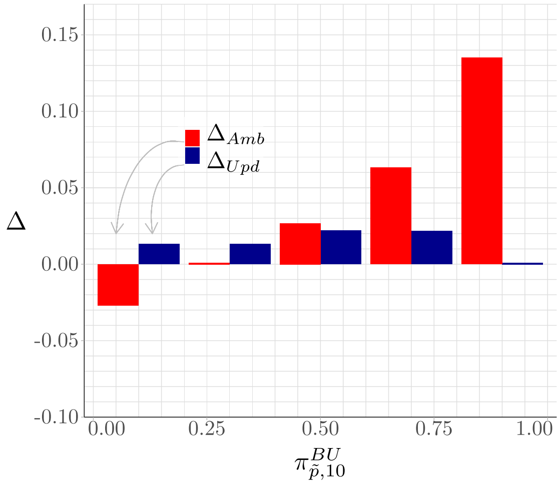

Fig. 4.6. Δ_{Amb} and Δ_{Upd} under the smooth model for five given $\pi^{BU}_{\tilde{p},10}$s. *Note.* The impact on MPs coming from changes in ambiguity aversion on learning is not dominated across the whole probability range by the impact due to divergence from Bayesian update.

of the null hypothesis that the regression line is the identity function ($p < 0.05$ for both S and L).[10] We now compare the impact that this divergence has on choice with the impact of changes in ambiguity attitudes.

Recall that, for a decision maker having observed a sample of size n' with elicited posterior probability over urn composition $\mu_{n'}$, the MP of the prospect $(x, E_{\tilde{p}})$ under the smooth ambiguity model is given by $M_{\tilde{p}}^{\mu_{n'},\phi_{n'}} = \phi_{n'}^{-1}\left(\int_{[0,1]} \phi_{n'}(p) d\mu_{n'}(p)\right)$. If her beliefs change according to Bayesian update – so $\mu_{n'} = \mu_{n'}^{BU}$ – and her ambiguity attitude transformation function is invariant across sample sizes – so $\phi_{n'} = \phi_n$ for another sample size n – then her matching probability $M_{\tilde{p}}^{\mu_{n'},\phi_{n'}} = M_{\tilde{p}}^{\mu_{n'}^{BU},\phi_n}$. The divergence from this invariant-ambiguity-attitude Bayesian-update benchmark can be decomposed as follows:

$$M_{\tilde{p}}^{\mu_{n'},\phi_{n'}} - M_{\tilde{p}}^{\mu_{n'}^{BU},\phi_n} = \underbrace{M_{\tilde{p}}^{\mu_{n'},\phi_{n'}} - M_{\tilde{p}}^{\mu_{n'},\phi_n}}_{\Delta_{\text{Amb}}} + \underbrace{M_{\tilde{p}}^{\mu_{n'},\phi_n} - M_{\tilde{p}}^{\mu_{n'}^{BU},\phi_n}}_{\Delta_{\text{Upd}}} \tag{8}$$

The first term, Δ_{Amb}, gives the difference in MPs due to the change in ambiguity aversion – as reflected in the transformation function – between n' and n, under fixed (elicited posterior) beliefs $\mu_{n'}$. The second term, Δ_{Upd}, reflects the MP-based divergence from Bayesian update, with fixed transformation function ϕ_n. We calculate Δ_{Amb} and Δ_{Upd} for $n = 5$, $n' = 10$ both at the aggregate and at the individual levels using aggregate and individual estimates of the transformation function and posterior beliefs, respectively.

Fig. 4.6 plots the values of Δ_{Amb} and Δ_{Upd} inferred from aggregate estimates of elicited posterior (second-order) beliefs (Appendix B.1) and the transformation functions ϕ_5, ϕ_{10} (see Table C.4, Appendix C), for different levels of the mean Bayesian update probability $\pi^{BU}_{\tilde{p},10}$. The impact on MPs coming from changes in ambiguity aversion on learning is not dominated across the whole probability range by the impact due to divergence from Bayesian update. Indeed, except for $\pi^{BU}_{\tilde{p},10} = 0.3$ where $|\Delta_{\text{Amb}}|$ is clearly smaller than $|\Delta_{\text{Upd}}|$, the ratio of $|\Delta_{\text{Amb}}|$ to the total absolute discrepancy $|\Delta_{\text{Amb}}| + |\Delta_{\text{Upd}}|$ is greater than 0.5, and of the order of 0.7 or higher for unlikely and likely winning events, i.e. $\pi^{BU}_{\tilde{p},10} = 0.1,\ 0.7,\ 0.9$ (Table C.4, Appendix C).

We also calculate Δ_{Amb} and Δ_{Upd} using individual-level elicited beliefs and transformation functions, for three categories of likelihood: unlikely ($\pi^{BU}_{\tilde{p},10} \le 0.2$; 75 observations), moderately likely ($\pi^{BU}_{\tilde{p},10} \in [0.4, 0.6]$; 124 observations), and likely ($\pi^{BU}_{\tilde{p},10} \ge 0.8$; 75 observations).[11] For the three categories of likelihood of the winning event, our observations do not conform with equal $|\Delta_{\text{Amb}}|$ and $|\Delta_{\text{Upd}}|$. Specifically, 64 out of 75, 109 out of 124, and 64 out of 75 observations satisfy $|\Delta_{\text{Amb}}| > |\Delta_{\text{Upd}}|$ for unlikely ($\pi^{BU}_{\tilde{p},10} \le 0.2$), moderately likely ($\pi^{BU}_{\tilde{p},10} \in [0.4, 0.6]$), and likely ($\pi^{BU}_{\tilde{p},10} \ge 0.8$) events, respectively (sign-tests: $p < 0.001$).

These analyses indicate that, for small sample sizes, ambiguity attitude change with learning has an impact on willingness to bet which is not significantly and globally dominated by the impact of divergences from Bayesian update. If anything, the effect of ambiguity aversion change is more important, especially for unlikely and medium-to-likely winning events.[12]

[10] See Appendix B.1 for further details on elicited posteriors and their relationship to Bayesian update beliefs.

[11] To illustrate this categorization of observations, a subject that has sampled one red and nine black chips – and hence for which $\pi^{BU}_{red,10} \le 0.2$ – will have her MP for red classified in the category $\pi^{BU}_{\tilde{p},10} \le 0.2$ and her MP for black in the category $\pi^{BU}_{\tilde{p},10} \ge 0.8$. By contrast, the MPs for red and black of a subject that has sampled 5 red and 5 black would be classified in the category $\pi^{BU}_{\tilde{p},10} \in [0.4, 0.6]$.

[12] Whilst the definition of Δ_{Upd} uses the result of Bayesian update applied to an uninformative prior, we could have alternatively taken as a benchmark Bayesian update applied to the elicited subjective probability with no sampling. Appendix C (Figure C.1) contains this analysis, showing that it yields the same qualitative conclusions concerning the comparison of the impacts of ambiguity attitude change and divergences from Bayesian update.

Large sample sizes. We now briefly report the results from study L, which concentrates on large sample sizes. In aggregate-level analysis under the smooth ambiguity model, a likelihood ratio test fails to reject the model with a single transformation function for both sample sizes in study L ($p = 0.24$). Hence, in contrast with the findings at small sample sizes, our data provides virtually no systematic evidence for significant differences in the ambiguity-attitude transformation function when learning from large sample sizes.

However, it does suggest non-neutral ambiguity attitudes in all treatments, including at very large sample sizes ($n = 10000$). Recall that ambiguity neutrality corresponds to linear ϕ_n, i.e. the limit as $\theta_n \to 0$ with $\rho_n = 1$. So the better fit under the expo-power specification as compared to the power and exponential specifications (Appendix A.2) suggests ambiguity non-neutrality. This is confirmed at the individual level: the number of positive θ_ns is 56 (41) out of 60 for $n = 100$ ($n = 10000$). The corresponding binomial tests are each significant at the level of 1%.

5. Discussion

5.1. The evolution of ambiguity perception and attitudes through learning

Our studies had an eye to *breadth*, *precision* and *robustness*. *Breadth*: our experiments scan the range of learning situations – from no sampling through small and medium-sized samples (5, 10 draws) to large and very large (100, 10000 draws) sample sizes. *Precision*: by independently separating out beliefs, through explicit elicitation in tandem with preferences, our analyses can identify the impact of learning on ambiguity attitude in isolation from its consequences for probabilistic beliefs. *Robustness*: by employing standalone choice-based elicitation of beliefs and matching probabilities in a learning context, we gauge the robustness of our central findings, be it under "raw data" approaches as well as in terms of the smooth ambiguity model. We find that ambiguity attitudes are impacted by learning, with a move towards ambiguity neutrality as the sample size increases. Under the smooth ambiguity model, the behavioral impact of these effects is of a magnitude comparable to, if not higher than, the behavioral impact of deviations from Bayesian update of probabilistic beliefs.

Considering each sample size in our setup to correspond to a source of uncertainty, our finding of an impact of learning on ambiguity attitude can be read in terms of source dependence – and as such, is perfectly coherent with the insight behind decision models involving it (Marinacci 2015; Ergin and Gul 2009; Tversky and Fox 1995; Chew and Sagi 2008; Abdellaoui et al. 2011a). Whilst source dependence has been documented in the literature (Anantanasuwong et al., 2019; Abdellaoui et al., 2011a), the source dependence of the transformation function in the smooth model remains an open empirical question. The current paper is the first study, to our knowledge, to tackle it via independent econometric estimation of the model components.

The comparative behavioral importance of ambiguity attitude change relative to deviations from Bayesian learning suggests the interest of further research in this direction. For instance, financial crises are economic situations involving important changes in the quantity of information, with global capital flows before and after crises having been related to changes in information or ambiguity (Rey, 2015; Giannetti and Laeven, 2016). Though such cases are typically modeled in terms of exogenous information shocks, decision-theoretic approaches often invoke violations of Bayesian update (Hill, 2022). Our research motivates an interesting question about the potential role of ambiguity attitude change over and above (non-Bayesian) belief change in such situations.

5.2. "Vanishing ambiguity" and the long run

Our other notable finding – of residual ambiguity non-neutrality even for very large sample sizes – speaks to the question of long-run learning, and connects into a strand of the theoretical literature on the long-run behavior of ambiguity models, which has identified conditions under which they tend to standard preferences under risk (Marinacci, 1999, 2002; Maccheroni and Marinacci, 2005; Epstein and Schneider, 2003, 2007).

The $n = 100, 10000$ treatments in our study L map directly into the sorts of cases considered by this literature. They are consistent with the aforementioned "vanishing ambiguity" results (Marinacci, 2002; Epstein and Schneider, 2007),[13] to the extent that they find that ambiguity attitudes "move towards ambiguity neutrality" on learning. However, the ambiguity attitudes in the highest sampling treatment ($n = 10000$) never quite reach ambiguity neutrality: there remains a very small but significant gap. This could be because the sample size explored is not high enough.[14] It could also be due to the computer-based implementation of sampling in study L: although all tasks in that study, including the risky prospects, were fully computer-based (Section 3.4), it is possible that specific distrust of sampling may be behind the observed gap between large-sample ambiguity attitudes and risk. A third possibility is that there is a residual gap between decision making on the basis of very large amounts of information and decision making on the basis of given probabilities (i.e. decision making under risk). This may connect into the psychological literature suggesting a difference between "experience-based decision making" – where subjects decide how much to sample, though they typically sample significantly less than 100 times – and "description-based decision making" – which corresponds to choice under risk (Abdellaoui et al. 2011b; Glöckner et al. 2016 and references therein).[15]

[13] In the cited papers, "vanishing ambiguity" refers to convergence to the special cases of the models of interest (maximin EU; Choquet EU) that apply to decision under risk, namely standard SEU. In this paper, where SEU is not assumed under risk, this naturally corresponds to ambiguity neutrality.

[14] Note however that after a certain point, ambiguity attitudes change very slowly: our model-based analysis fails to find a significant difference in ambiguity attitude between the $n = 100$ treatment and the $n = 10000$ treatment.

[15] Note that, like our study L, all prospects and sampling in this literature are typically computer-based.

5.3. Methodological contributions and related experimental literature

A key experimental innovation in our approach is the conjoint use made of MPs and choice-based elicited beliefs to econometrically estimate the smooth ambiguity model. By contrast, the few recent estimation attempts systematically use the outcome scale to compare prospects, e.g. Cubitt et al. (2018); Berger and Bosetti (2020) used certainty equivalents (CEs), estimating both utility for risk ($u(.)$) and utility for ambiguity ($\phi[u(.)]$). Our MP-based evaluation of ambiguous prospects allows for a parsimonious estimation of the transformation function ϕ without requiring estimation of utility. In particular, since it does not rely on assumptions about EU holding for risky prospects, it is not subject to biases resulting from the usual discrepancies for EU under risk. Moreover, none of the previous studies explicitly and separately elicits beliefs.

A further experimental innovation is the strict control of the amount of learning via fixed-length, typically real-time sampling opportunities prior to choice. This contrasts with the psychological literature on the experience-description gap (e.g. Hertwig, 2004), as well as Ert and Trautmann (2014), which allow subjects to decide how much to sample. It also contrasts with recent work on hard-to-interpret signals, uncovering attitudes to the ambiguity of signals (Epstein and Halevy, 2024). A potential direction for future research would be to see how our results extend to situations of learning with differently sized samples of potentially ambiguous signals.

Assuming the neo-additive capacity model (Section 2.2), Baillon et al. (2018) estimate ambiguity attitudes from CEs of bets on NYSE stock returns. Comparing reactions to receiving different amounts of information, their central finding differs considerably from ours: whilst they find a shift in likelihood insensitivity with increased information, they find little evidence for an effect of information on their ambiguity aversion parameter. Experimental design factors could potentially explain this difference. Firstly, whilst they use differently-sized blocks of information (defined in terms of weeks of past data) about real stocks that subjects may have learnt about elsewhere, our treatments differ by the sample size observed, arguably providing more control over the information subjects possess. Secondly, their results rely on a concurrent parametric estimation yielding subjects' utility and complementarity ambiguity aversion index from CEs, adding another source of noise; by contrast, our MP elicitation provides this index immediately (Sections 2.2 and 4.1). Finally, beyond their assumed neo-additive capacity model, this index reflects some combination of ambiguity aversion and beliefs (Section 2.2), complicating interpretation of their results. Our analyses in terms of the smooth ambiguity model show that our conclusions are robust to such issues.

CRediT authorship contribution statement

M. Abdellaoui: Funding acquisition, Visualization, Conceptualization, Methodology, Writing – review & editing. **B. Hill:** Funding acquisition, Conceptualization, Methodology, Writing – review & editing. **E. Kemel:** Conceptualization, Methodology, Software, Visualization, Data curation, Data analysis, Review & editing. **H. Maafi:** Conceptualization, Methodology, Data collection, Review & editing.

Declaration of competing interest

No one.

Appendix. Supplementary material

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jet.2025.106093.

Data availability

Data sets are available on https://doi.org/10.3886/E238604V1.

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